



MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

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CEF-2113 – Algorithms and Data Structures Lab Laboratory Manual

Department	:	School of Basic Science, Humanities and Management			
Course Name & code	:	CEF 2113 & Algorithms and Data Structures Lab			
Semester & branch	:	III Sem & BTech Computer Science & Finance Technology			
Name of the faculty	:	Dr. Manjunatha			
No of contact hours/week:		L	T	P	C
		0	0	3	1

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Course Objectives ·

1. Design, develop, and implement programs using the concepts of Recursion and the mapping of 2D arrays to 1D arrays
2. Design, develop, and implement programs and applications using linear data structures such as Stacks, Queues, and Linked Lists
3. Design, Develop, and Implement programs and applications using the non-Linear Data structures such as trees and graphs
4. Application of different algorithm design techniques .

Evaluation plan : (Tentative)

- Internal Assessment Marks: 60%
- End-semester assessment of 2-hour duration: 40 %

Evaluation pattern

Internal Marks – 60 + End Sem - 40	
Internal Marks	Internal Assessment – 40 + Mid-sem - 20
Internal Assessment	Lab observations – 2*11(22) + Viva (18)
Lab Observation	Record (1*11) + Execution (1*11)
Viva	Quiz(18)

INSTRUCTIONS TO THE STUDENTS

Pre-Lab Session Instructions

- Be on time, adhere to the institution's rules, and maintain decorum.
- Leave your mobile phones, pen drives, and other electronic devices in your bag and keep the bag in the designated place in the lab.
- Must Sign in to the log register provided.
- Make sure to occupy the allotted system and answer the attendance.

In-Lab Session Instructions

- Follow the instructions on the allotted exercises.
- Show the program and results to the instructors on completion of experiments.
- Copy the program and results for the lab record.
- Prescribed textbooks and class notes can be kept ready for reference if required.

General Instructions for the Exercises in Lab

- Academic honesty is required in all your work. You must solve all programming assignments independently, except where group work is authorized. This means you must not take, show, give, or otherwise allow others to take your program code, problem solutions, or other work.
- The programs should meet the following criteria:
 - Programs should be interactive with appropriate prompt messages, error messages if any, and descriptive messages for outputs.
 - Programs should perform input validation (Data type, range error, etc.), give appropriate

error messages, and suggest corrective actions.

- Comments should be used to give the statement of the problem, and every function should indicate the purpose of the function, inputs, and outputs.
- Statements within the program should be properly indented.
- Use meaningful names for variables and functions.
- Make use of constants and type definitions wherever needed.
- The exercises for each week are divided into three sets:
 - Solved solutions
 - Lab exercises - to be completed during lab hours
 - Additional Exercises - to be completed outside the lab or in the lab to enhance the skill

Questions for lab tests and examinations are not necessarily limited to the questions in the manual but may involve some variations and/or combinations of the questions.

THE STUDENTS SHOULD NOT

- Possess mobile phones or any other electronic gadgets during lab hours.
- Go out of the lab without permission.
- **Change/update any configuration in your allotted system. If so, the student will lose internal assessment marks**

Week 1 - Mapping of 2-D Arrays to 1-D Arrays in Java

Learning Objectives

After completing this experiment, students will be able to:

1. Understand the memory representation of 2-D arrays.
2. Convert a 2-D array into a 1-D array using Row-Major Mapping.
3. Convert a 2-D array into a 1-D array using Column-Major Mapping.
4. Access elements using mapping formulas.
5. Apply array mapping concepts in data structure applications.

Two-Dimensional Array

A two-dimensional array consists of rows and columns.

Example:

```
1 2 3
4 5 6
7 8 9
```

Declaration:

```
int[][] arr =
{
    {1,2,3},
    {4,5,6},
    {7,8,9}
};
```

Why Mapping is Required?

Computer memory is linear in nature. Therefore, multidimensional arrays are stored internally as one-dimensional sequences.

Mapping converts a 2-D array into a 1-D array representation.

Example:

2-D Array

```
1 2 3
4 5 6
7 8 9
```

1-D Representation

1 2 3 4 5 6 7 8 9

Row-Major Mapping

In Row-Major Mapping, elements are stored row by row.

Formula

For an element at position:

$$A[i][j]$$

The corresponding index in the 1-D array is:

$$\text{Index} = i \times \text{Number_of_Columns} + j$$

Example

For

1 2 3

4 5 6

7 8 9

Mapping becomes:

1 2 3 4 5 6 7 8 9

Column-Major Mapping

Formula

$$\text{Index} = j \times \text{Number_of_Rows} + i$$

Example

For

1 2 3

4 5 6

7 8 9

Mapping becomes:

1 4 7 2 5 8 3 6 9

Algorithm (Row-Major Mapping)

1. Read the number of rows and columns.
2. Create a 2-D array.
3. Accept elements from the user.
4. Create a 1-D array of size rows $\times$ columns.
5. Traverse the 2-D array row-wise.
6. Store elements into the 1-D array.
7. Display the mapped array.

Sample Program

Program: Convert 2-D Array into 1-D Array Using Row-Major Mapping

```
import java.util.Scanner;

public class RowMajorMapping
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();
        System.out.print("Enter number of columns: ");
        int cols = sc.nextInt();
        int[][] arr = new int[rows][cols];
        int[] oneD = new int[rows * cols];
        System.out.println("Enter array elements:");
        for(int i = 0; i < rows; i++)
        {
            for(int j = 0; j < cols; j++)
            {
                arr[i][j] = sc.nextInt();
            }
        }
    }
}
```

```
int k = 0;
for(int i = 0; i < rows; i++)
{
    for(int j = 0; j < cols; j++)
    {
        oneD[k++] = arr[i][j];
    }
}
System.out.println("Mapped 1-D Array:");
for(int value : oneD)
{
    System.out.print(value + " ");
}
}
```

Sample Output

Enter number of rows: 3

Enter number of columns: 3

Enter array elements:

1 2 3

4 5 6

7 8 9

Mapped 1-D Array:

1 2 3 4 5 6 7 8 9

Week 1: Exercises:

1. Write a Java program to convert a 2-D array into a 1-D array using Column-Major Mapping.

Expected Output

Original Array

1 2 3

4 5 6

7 8 9

Column Major Mapping

1 4 7 2 5 8 3 6 9

2. Write a Java program to calculate the 1-D index position of an element present at row i and column j using the Row-Major Mapping formula.

Example

Rows = 4

Columns = 5

Element Position = (2,3)

Index = $2 \times 5 + 3$

Index = 13

Expected Output

Mapped Index = 13

3. Write a Java program to convert a given 1-D array back into a 2-D array of specified dimensions.

Example Input

1-D Array:

10 20 30 40 50 60

Rows = 2

Columns = 3

Expected Output

10 20 30

40 50 60

4. Store two matrices in 1-D arrays using Row-Major Mapping and perform matrix addition.

Example

Matrix A

1 2

3 4

Matrix B

5 6

7 8

Expected Output

Sum Matrix

6 8

10 12

5. Write a Java program to read a matrix and determine whether it is a sparse matrix (contains more zeros than non-zero elements). Store all non-zero elements in a 1-D array using the format: row column value

Example Input

0 0 3

0 5 0

7 0 0

Expected Output

Sparse Matrix Representation

0 2 3

1 1 5

2 0 7

6. Write a Java program to find an element in a 2-D array and display its corresponding 1-D mapped index using Row-Major Mapping.

Example Input

Matrix

10 20 30

40 50 60

70 80 90

Search Element: 60

Expected Output

Element Found at:

Row = 1

Column = 2

Mapped Index = 5

Formula Used

Index = (Row × Number_of_Columns) + Column

Additional Exercises:

7. Write a Java program to:
 - Read a matrix.
 - Store it in a 1-D array using Row-Major Mapping.
 - Generate the transpose of the matrix.
 - Display the transposed matrix.
8. Write a Java program to:
 - Read two matrices.
 - Store both matrices as 1-D arrays using Row-Major Mapping.
 - Perform matrix multiplication.
 - Display the resultant matrix.

Week-2: Solving Problems Using Recursion in Java

Aim: To understand the concept of recursion and develop Java programs to solve computational problems using recursive techniques.

Learning Objectives

After completing this laboratory, students will be able to:

- Understand recursive problem-solving techniques.
- Differentiate between recursion and iteration.
- Implement recursive functions in Java.
- Analyze recursive calls and execution flow.
- Solve mathematical and data structure problems using recursion.

Prerequisites

Students should be familiar with:

- Java methods
- Conditional statements
- Loops
- Function calls
- Basic mathematical computations

What is Recursion?

Recursion is a programming technique in which a method calls itself repeatedly until a terminating condition is reached.

A recursive program consists of:

1. Base Case

Stops the recursion.

2. Recursive Case

Calls itself with a smaller or simpler problem.

General Structure

```
returnType function(parameters)
{
    if(base_condition)
        return value;

    return function(smaller_problem);
}
```

Why Recursion?

Recursion is useful for:

- Tree traversal

- Graph traversal
- Divide and Conquer algorithms
- Backtracking problems
- Mathematical computations

Examples:

- Factorial
- Fibonacci Series
- Tower of Hanoi
- Binary Search
- Tree Traversal

Example: Recursive Factorial Algorithm

1. Read a number n.
2. If n is 0 or 1, return 1.
3. Otherwise return $n \times \text{factorial}(n-1)$.
4. Display the result.

Program: Factorial Using Recursion

```
import java.util.Scanner;
public class RecursiveFactorial
{
    static long factorial(int n)
    {
        if(n == 0 || n == 1)
            return 1;
        return n * factorial(n - 1);
    }

    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        System.out.println("Factorial = " + factorial(n));
    }
}
```

Sample Output

Enter a number: 5

2

Factorial = 120

Week 2: Exercises:

1. Write a Java program to generate Fibonacci numbers using recursion.
2. Write a Java program to find the sum of digits of a given number using recursion.
3. Write a Java program to reverse a given integer using recursion.
4. Write a Java program to find the GCD using Euclid's Recursive Algorithm.

Additional Exercises:

1. Write a Java program to search for an element in a sorted array using recursive binary search.
2. Write a Java program to compute x^n using recursion.

Week-3: Stacks and its Applications

Learning Objectives

At the end of this laboratory session, students will be able to:

1. Understand the Stack Abstract Data Type (ADT).
2. Implement stack operations using arrays and Java collections.
3. Apply stack concepts to solve computational problems.
4. Develop programs using stack-based algorithms.
5. Analyze the time complexity of stack operations.
6. Use stacks in expression evaluation and string processing applications.

Stack

A **Stack** is a linear data structure that follows the **LIFO (Last In First Out)** principle.

Basic Operations

- **push()** : Insert an element at the top.
- **pop()** : Remove the top element.
- **peek()** : View top element.
- **isEmpty()** : Check whether stack is empty.
- **size()** : Number of elements.

Applications of Stack

- Function call management
- Expression conversion
- Expression evaluation
- Parentheses matching
- Undo operations
- Browser navigation
- Backtracking algorithms
- Depth First Search (DFS)

Sample Exercise

Program : Implementation of Stack Using Array

Algorithm

1. Create a stack using an array.
2. Initialize top = -1.
3. Display a menu with Stack operations.
4. Accept user's choice.
5. Perform Push, Pop, Peek, or Display operation.
6. Repeat until the user chooses Exit.

Program:

```
import java.util.Scanner;

class StackArray {
    int stack[];
    int top;

    StackArray(int size) {
        stack = new int[size];
        top = -1;
    }

    void push(int value) {
        if (top == stack.length - 1) {
            System.out.println("Stack Overflow");
        } else {
            stack[++top] = value;
            System.out.println(value + " inserted into stack.");
        }
    }

    void pop() {
        if (top == -1) {
            System.out.println("Stack Underflow");
        } else {
            System.out.println("Deleted Element: " + stack[top--]);
        }
    }

    void peek() {
        if (top == -1) {
            System.out.println("Stack is Empty");
        } else {
            System.out.println("Top Element: " + stack[top]);
        }
    }

    void display() {
        if (top == -1) {
            System.out.println("Stack is Empty");
        } else {
            System.out.println("\nStack Elements:");
            for (int i = top; i >= 0; i--) {
```

```

        System.out.println(stack[i]);
    }
}
}

```

```

public class StackDemo {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Stack Size: ");
        int size = sc.nextInt();

        StackArray s = new StackArray(size);

        int choice, value;

        do {
            System.out.println("\n----- STACK MENU -----");
            System.out.println("1. Push");
            System.out.println("2. Pop");
            System.out.println("3. Peek");
            System.out.println("4. Display");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");

            choice = sc.nextInt();

            switch (choice) {

                case 1:
                    System.out.print("Enter value to push: ");
                    value = sc.nextInt();
                    s.push(value);
                    break;

                case 2:
                    s.pop();
                    break;

                case 3:
                    s.peek();

```

```

        break;

    case 4:
        s.display();
        break;

    case 5:
        System.out.println("Exiting Program...");
        break;

    default:
        System.out.println("Invalid Choice!");
    }

} while (choice != 5);

sc.close();
}
}

```

Output:

Enter Stack Size: 5

----- STACK MENU -----

1. Push
2. Pop
3. Peek
4. Display
5. Exit

Enter your choice: 1

Enter value to push: 10

10 inserted into stack.

Enter your choice: 1

Enter value to push: 20

20 inserted into stack.

Enter your choice: 1

Enter value to push: 30

30 inserted into stack.

Enter your choice: 4

Stack Elements:

30

20

10

Enter your choice: 3

Top Element: 30

Enter your choice: 2

Deleted Element: 30

Enter your choice: 4

Stack Elements:

20

10

Enter your choice: 5

Exiting Program...

Week-3 Exercises:

1. Write a Java program that accepts a string from the user and reverses it using a stack.
2. Write a Java program that accepts an arithmetic expression containing the symbols (), {}, and [], and determines whether the expression is balanced using a stack.
3. Write a Java program that accepts an infix expression and converts it into an equivalent postfix expression using stack operations while considering operator precedence and associativity.
4. Write a Java program that accepts a postfix expression consisting of single-digit operands and arithmetic operators (+, -, *, /) and evaluates the expression using a stack.
5. Write a Java program that accepts a decimal number from the user and converts it into its binary representation using a stack.

Additional Exercises:

1. Write a Java program to implement two stacks using a single array. The program should provide separate operations for both stacks, including:
 - Push an element into Stack 1
 - Push an element into Stack 2

Pop an element from Stack 1
Pop an element from Stack 2
Display the contents of Stack 1
Display the contents of Stack 2
Check for overflow and underflow conditions

2. Given an array of integers, the Next Greater Element for an element is the first element on its right side that is greater than it. If no such element exists, the NGE is considered **-1**. Write a Java program to find the next greater element for each element in the array using a stack-based approach.

Input: [4, 5, 2, 25, 7, 8]

Output:

Element	Next Greater Element
4	5
5	25
2	25
25	-1
7	8
8	-1

Week-4: Queue and its Applications

Learning Outcomes

At the end of this laboratory session, students will be able to:

1. Understand the FIFO (First In First Out) principle of queues.
2. Implement queue operations using Java collections.
3. Apply queues to solve practical problems.
4. Analyze the time complexity of queue operations.
5. Develop problem-solving skills using queue-based approaches.

A **Queue** is a linear data structure that follows the **First In First Out (FIFO)** principle. The element inserted first will be removed first.

Common Queue Operations

- **Enqueue:** Insert an element at the rear.
- **Dequeue:** Remove an element from the front.
- **Peek/Front:** View the first element.
- **isEmpty():** Check whether the queue is empty.
- **Size():** Determine the number of elements.

Applications of Queues

- CPU Scheduling
- Printer Spooling
- Task Scheduling
- Network Packet Routing
- Ticket Reservation Systems
- Customer Service Systems
- Breadth First Search (BFS)
- Message Queues

Sample Program:

Queue Implementation Using Array

1. Initialize front = 0 and rear = -1.
2. For enqueue:
 - Check whether the queue is full.
 - Increment rear.
 - Insert the element.
3. For dequeue:
 - Check whether the queue is empty.
 - Remove the element at front.
 - Increment front.
4. Display the elements between front and rear.

Java Code:

```
import java.util.Scanner;
class ArrayQueue {

    private int[] queue;
    private int front;
    private int rear;
    private int capacity;

    ArrayQueue(int size) {
        capacity = size;
        queue = new int[capacity];
        front = 0;
        rear = -1;
    }

    void enqueue(int value) {

        if (rear == capacity - 1) {
            System.out.println("Queue Overflow");
            return;
        }

        queue[++rear] = value;
    }

    int dequeue() {

        if (front > rear) {
            System.out.println("Queue Underflow");
            return -1;
        }

        return queue[front++];
    }

    void display() {

        if (front > rear) {
            System.out.println("Queue is empty");
            return;
        }
    }
}
```

```

        System.out.print("Queue: ");

        for (int i = front; i <= rear; i++)
            System.out.print(queue[i] + " ");

        System.out.println();
    }
}

public class QueueDemo {

    public static void main(String[] args) {
        ArrayQueue q = new ArrayQueue(5);
        q.enqueue(10);
        q.enqueue(20);
        q.enqueue(30);
        q.enqueue(40);
        q.display();
        System.out.println("Deleted: " + q.dequeue());
        q.display();
    }
}

```

Sample Output

Queue: 10 20 30 40

Deleted: 10

Queue: 20 30 40

Week 3 Exercises:

1. Write a Java program to implement a Queue using an array and demonstrate its application in a Printer Job Scheduling System. In the printer system, print jobs are processed in the order they are received (FIFO).
The program should support the following operations:
 - Add a print job to the queue (Enqueue).
 - Process a print job (Dequeue).
 - Display the next print job to be processed.
 - Display all print jobs currently waiting in the queue.
 - Handle Queue Overflow and Queue Underflow conditions.
2. Write a Java program to implement a Circular Queue using an array and perform enqueue, dequeue, peek, and display operations. Demonstrate how circular queues overcome the limitations of a linear queue by reusing vacant spaces created after deletions.

3. Write a Java program to implement a Double-Ended Queue (Deque) using an array. The program should support the following operations:
 - Insert an element at the front of the deque.
 - Insert an element at the rear of the deque.
 - Delete an element from the front of the deque.
 - Delete an element from the rear of the deque.
 - Display all elements present in the deque.
 - Handle Overflow and Underflow conditions.
4. Write a Java program to implement a Priority Queue. Perform insertion, deletion, and display operations, ensuring that elements with higher priority are processed before elements with lower priority. Demonstrate the working of the priority queue using suitable test cases.

Additional Exercises:

5. Given an integer array and a window of size k , develop an efficient Java program using a **Deque** to determine the maximum element in every window.

Example:

Input:

Array = [1, 3, -1, -3, 5, 3, 6, 7]

$k = 3$

Expected output:

3 3 5 5 6 7

6. Design and implement a Java program for a task scheduler in which every task has:

Task ID
Task Name
Priority
Arrival Time
Execution Time

The scheduler should:

- Process higher-priority tasks first.
- Process tasks with the same priority according to arrival order.
- Maintain waiting tasks using an appropriate queue.
- Display the final execution sequence.

Week-5 Single Linked List

Learning Outcomes:

Upon successful completion of this laboratory session, students will be able to:

- Implement a singly linked list using Java.
- Perform insertion, deletion, and traversal operations.
- Analyze the advantages of linked lists over arrays.
- Apply linked lists to solve real-world problems.
- Develop efficient solutions using linked list operations.

A Singly Linked List (SLL) is a dynamic linear data structure consisting of nodes. Each node contains:

- Data
- Reference (link) to the next node

Unlike arrays, linked lists do not require contiguous memory locations and can grow or shrink dynamically during runtime.

Structure of a Node:

```
+-----+-----+
| Data | Next |
+-----+-----+
```

Eg: $10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow \text{NULL}$

Advantages of Singly Linked List

- Dynamic size allocation.
- Efficient insertion and deletion.
- No memory wastage due to fixed size.
- Ease of implementation of stacks and queues.

Basic Operations:

1. Insertion

- Insert at beginning
- Insert at end
- Insert at a specific position

2. Deletion

- Delete from beginning
- Delete from the end
- Delete a given node

3. Traversal

- Visit every node in sequence.

4. Searching

- Find a specified element.

Sample Exercise:

Write a Java program to implement a Singly Linked List and perform the following operations:

1. Insert elements at the end.
2. Display all elements.
3. Search for an element entered by the user.

Java Code”

```
import java.util.Scanner;
class Node {
    int data;
    Node next;

    Node(int data) {
        this.data = data;
        this.next = null;
    }
}

public class SinglyLinkedList {

    Node head;
    void insert(int data)
    {
        Node newNode = new Node(data);

        if (head == null) {
            head = newNode;
            return;
        }

        Node temp = head;

        while (temp.next != null) {
            temp = temp.next;
        }

        temp.next = newNode;
    }

    void display() {
        Node temp = head;
```

```

        while (temp != null) {
            System.out.print(temp.data + " -> ");
            temp = temp.next;
        }

        System.out.println("NULL");
    }

    boolean search(int key) {
        Node temp = head;

        while (temp != null) {
            if (temp.data == key)
                return true;

            temp = temp.next;
        }

        return false;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        SinglyLinkedList list = new SinglyLinkedList();
        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();
        for(int i=0;i<n;i++) {
            list.insert(sc.nextInt());
        }

        System.out.println("Linked List:");
        list.display();
        System.out.print("Enter element to search: ");
        int key = sc.nextInt();
        if(list.search(key))
            System.out.println("Element found.");
        else
            System.out.println("Element not found.");
    }
}

```

Week-5 Exercises:

1. Write a Java program to implement a Singly Linked List and insert elements at the beginning. Display the resulting list after each insertion.
2. Write a Java program to delete a specific element from a singly linked list and display the updated list.
3. Write a Java program to count and display the total number of nodes present in a singly linked list.
4. Write a Java program to traverse a singly linked list and find the maximum element stored in the list.
5. Write a Java program to reverse a singly linked list without using an additional linked list.
6. Write a Java program to search for a given element in a singly linked list and display whether it is found or not.
7. Write a Java program to insert a new node after a specified value in a singly linked list.
8. Write a Java program to find the middle node of a singly linked list using the slow and fast pointer technique.

Additional Exercises:

9. Create a singly linked list where each node stores:
Student ID
Student Name
CGPA

Implement operations to:

- Insert student records
- Display all records
- Search a student by ID
- Delete a student record